

A Theory of Slack

How Economic Slack Shapes Markets, Business Cycles, and Policies

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Draft version: May 2026

Draft URL: pascalmichailat.org/18/

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CHAPTER 10.

Market efficiency and inefficiency

We have now built a slackish model of markets, and studied its positive properties: what are the price, slack, and output prevailing in the market, and how these variables respond to supply and demand shocks. In this chapter we turn to the normative, welfare-related properties of the model: What is the efficient market allocation? Can we expect the market to operate efficiently? If not, how far from efficiency might the market be?

What do we mean by efficient? The efficient allocation is the allocation that maximizes the social welfare generated by the market. Knowing markets' efficient allocations is critical for designing the best possible stabilization policies, as we will see in part V.

10.1. Overview of the sufficient-statistic approach

Before we begin we must decide which model to use for the welfare analysis. We have introduced a range of slackish models: a static model with exogenous capacity; a static model with endogenous capacity; a dynamic model with exogenous capacity, which itself could be extended with endogenous capacity. We have also seen that these models can be built around a number of matching functions—as long as they satisfy a few assumptions—and a wide array of price norms. Which one should we pick?

It turns out that we do not need to answer this question, because we adopt the sufficient-statistic approach to welfare analysis. We simply solve the problem of a market planner who is trying to maximize welfare in a more general framework. Typically, being more general adds complexity. Here, however, the generalization cleans things up and highlights

the mechanisms and forces at play better.

Traditionally, welfare analysis is structural. This approach comes with two limitations that the sufficient-statistic approach resolves. First, the structural approach assumes the entire structure of the market and studies welfare and policy within this structure. The limitation is that the market structure is assumed religiously but largely determines the policy conclusions. The policy insights are in many ways baked into the initial assumptions about model structure. Second, the structural approach requires calibrating all the parameters in the model even though many of these parameters are generally not observable in the real world.

With the sufficient-statistic approach, we do not assume the entire structure of the market. Instead, we make a minimal set of assumptions on the welfare function and market structure that allow for welfare analysis. This is beneficial because, unlike the structural approach, it allows the analysis to be applied to a range of models: any model that satisfies the minimal set of assumptions. Additionally, from a practical side, since we are making fewer assumptions, it is more likely that our analysis is valid in the real world.

At the end of the chapter, we will apply the general analysis to each of the models that we have introduced in the book, and we will map the sufficient-statistic results into structural results.

10.2. Market planner's problem

We consider a market planner who aims to allocate the goods supplied to the market efficiently. The market planner allocates some of the goods for consumption, some for matching, and leaves some of them unsold. The planner's objective is to maximize market welfare. What should the planner do?

10.2.1. Market slack

We assume that there is slack on the market, so a share $u \in (0, 1)$ of all goods available on the market are unsold. The rest of the goods are sold to buyers.

10.2.2. Beveridge curve

We also assume that the market features a Beveridge curve. This means that the visit rate is a strictly decreasing and convex function $v(u)$ of the slack rate. This is the sole structural assumption that we make in the welfare analysis. We do not need to assume a matching function or price norm; we only need the reduced-form Beveridge curve.

Why do visits enter the planner's problem? Because visits require resources, measured by the matching cost, $\kappa > 0$. The matching cost is the number of goods that have to be devoted to any one visit.

The Beveridge curve says that as the planner reduces the slack rate, the visit rate increases. So as the number of unsold goods falls, more goods must be allocated to visits and matching buyers and sellers. This is the central tradeoff that the planner must resolve: how to balance unsold goods against goods allocated to matching.

10.2.3. Market welfare

Market welfare measures the well-being generated by market activity. In our slackish market model, market welfare is not very complicated. It starts from the market capacity, which is k goods.

First, not all goods are sold. The slack rate is u , so only $(1 - u)k$ goods are sold to buyers. The uk goods that remain unsold have no social value.

However, not all $(1 - u)k$ sold goods are consumed. Some goods are instead devoted to matching and, as they are not consumed, they do not produce direct social value. Thus, we must net out the amount of goods devoted to matching: we need to subtract κvk goods from the $(1 - u)k$ goods sold to buyers. (Since v is the visit rate, vk is the number of visits in the market.)

Overall, $(1 - u)k - \kappa vk$ goods are consumed through market purchases, so market welfare is

$$(10.1) \quad [1 - u - \kappa v] k.$$

10.2.4. Planner's problem

We are now in a position to state the market planner's problem. We begin by plugging the Beveridge curve into market welfare, to write market welfare as a function of the slack rate:

$$(10.2) \quad \mathcal{W}(u) = [1 - u - \kappa v(u)] k.$$

The market planner chooses a slack rate $u \in (0, 1)$ to maximize market welfare $\mathcal{W}(u)$. The slack rate u^* that maximizes market welfare is the efficient slack rate. The efficient slack rate is the best slack rate from a social perspective.

10.3. Solution to the planner's problem

We now determine the efficient slack rate, u^* . Ideally the planner would prefer to have no unsold goods and no goods allocated to matching, but that is not feasible in a slackish market. The Beveridge curve imposes that some goods are always unsold, and some visits are always required so some buyers and sellers match.

10.3.1. Analytical solution

We start by analytically determining the efficient slack rate, u^* . Because the Beveridge curve $v(u)$ is strictly convex, we see in (10.2) that the welfare function $\mathcal{W}(u)$ is strictly concave. Indeed, $-v(u)\kappa k$ is the opposite of a strictly convex function and thus strictly concave, and adding the linear term $[1 - u]k$ preserves strict concavity (result G.17). Because the welfare function is strictly concave, the first-order condition is sufficient to find the unique maximum of the welfare function (result G.21).

To find the efficient slack rate, we simply set the derivative of the welfare function to zero: $\mathcal{W}'(u) = 0$. The efficient slack rate satisfies

$$0 = -[1 + \kappa v'(u)]k.$$

This means that the efficient slack rate is implicitly defined by

$$(10.3) \quad v'(u) = -\frac{1}{\kappa}.$$

Efficiency requires the slope of the Beveridge curve equals $-1/\kappa$. There is a tradeoff between too much slack and too little slack, both of which are undesirable. The tradeoff is modulated by the Beveridge curve. The efficient slack rate guarantees that there is some slack to keep resources devoted to matching low, but not too much to keep the number of unsold goods low.

Formally, formula (10.3) says that when the market operates efficiently, the welfare costs and benefits from moving one good from sold to unsold are equalized. When one good is not sold anymore, consumption drops by 1 good. Having one more good unsold also means having $-v'(u) > 0$ fewer visits. Each visit reduces consumption by the matching cost, κ , so consumption increases by $-v'(u)\kappa > 0$ goods. When costs and benefits are equalized, we have $1 = -v'(u)\kappa$, which is equivalent to (10.3).

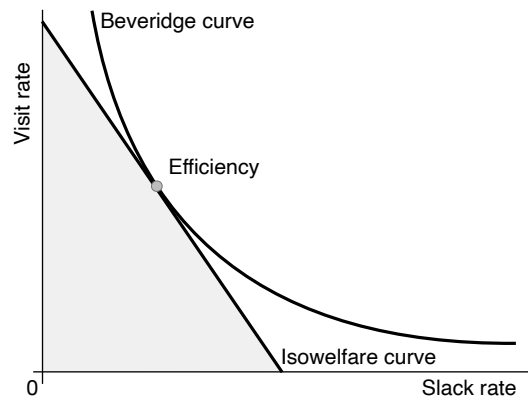
10.3.2. Graphical solution

Next, we graphically determine the efficient slack rate (figure 10.1A). To do that, we introduce isowelfare curves:

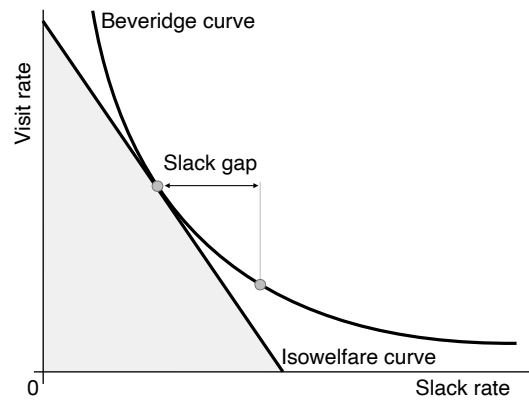
$$(10.4) \quad v = A - \frac{1}{\kappa} \cdot u,$$

where $A > 0$ governs the welfare level along the curve. On any isowelfare curve, the visit rate and slack rate are such that social welfare does not change.

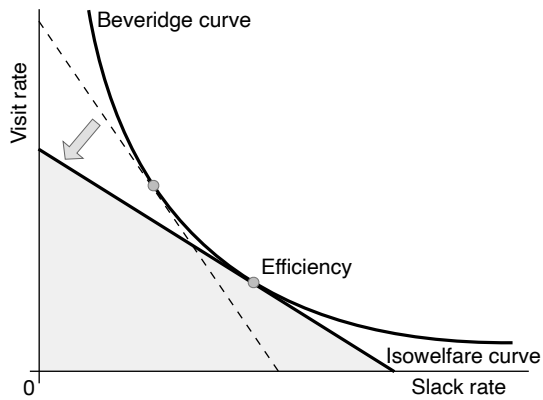
Suppose that the planner could place the market on any isowelfare curve. Which one would they choose? Well, the planner would want to be on an isowelfare curve as close



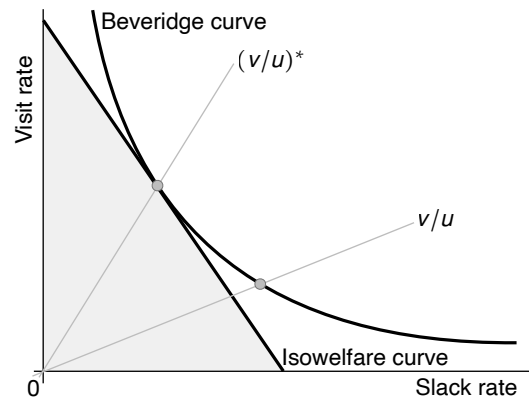
A. Efficient good allocation



B. Inefficient good allocation: excessive slack



C. Comparative statics: higher matching cost



D. Efficient and inefficient $v-u$ ratios

FIGURE 10.1. Efficiency and inefficiency in slackish market model

The isowelfare line is given by (10.4). The shaded area indicates all good allocations with higher welfare than the isowelfare line.

to the origin as possible. Points closer to the origin are characterized by fewer visits and fewer unsold goods. Visits and unsold goods are costly, so the points on isowelfare curves that lie closer to the origin have higher welfare than the points that are further out.

However, the isowelfare curves that are deep inside, close to the origin, do not cross the Beveridge curve: they are not feasible. The market planner must respect the Beveridge curve so must choose a point that is also on the Beveridge curve. Accordingly, the optimal choice for the planner is to pick the isowelfare curve that is as far in as possible while still touching the Beveridge curve: the isowelfare curve that is tangent to the Beveridge curve. The tangent isowelfare curve has the highest welfare while respecting the feasibility constraint represented by the Beveridge curve.

Then, the efficient allocation is the point of tangency between the Beveridge curve and isowelfare curve. Since the slope of the Beveridge curve is $v'(u)$ and the slope of the isowelfare curve is $-1/\kappa$, we recover the efficiency condition (10.3) with our graphical approach.

At the point of tangency, the market operates efficiently; at other points on the Beveridge curve, it operates inefficiently. At any point below the efficiency point, for instance, slack is inefficiently high (figure 10.1B). This situation generates a positive slack gap, $u - u^* > 0$. As the market becomes tighter and tighter, at some point it crosses efficiency. If the market keeps tightening, the slack gap becomes negative, that is, $u - u^* < 0$. In this case, the slack rate is below the efficient slack rate.

10.3.3. Comparative statics

From equation (10.3), we see what happens if the matching cost changes. Given that the Beveridge curve $v(u)$ is convex, $v'(u)$ is increasing in u . If the matching cost κ goes up, then $-1/\kappa$ falls, meaning that the Beveridge curve must be flatter at efficiency. As a result, the efficient slack rate u^* increases. If the matching cost goes up, that makes the concern about too many goods devoted to matching more important, which tilts the balance in the tradeoff towards allowing a bit more slack.

We can also perform the comparative statics on the efficiency diagram (figure 10.1C). If the matching cost κ goes up, the isowelfare curve (10.4) becomes flatter. A flatter line means that the new tangency point occurs further out. Thus, the new efficient slack rate is now higher.

10.4. Sufficient-statistic formula for the efficient v-u ratio

There are two potential issues with the efficiency formula (10.3) that make it hard to implement in practice. First, the formula defines the efficient slack rate or visit rate only implicitly. If the formula holds, then the slack and visit rates are efficient, but their values

are not given. Second, both in theory and in practice, the Beveridge curve is convex, so the slope of the Beveridge curve constantly changes with the tightness of the market. Measuring a slope in the data is hard enough: measuring the slope at any point in time to know if the condition is satisfied would be so hard as to be impractical.

To alleviate these issues, we reformulate the efficiency condition so that it involves model variables explicitly and so that it only involves stable statistics (unlike the Beveridge slope). We are looking for a formula in which we can plug market statistics that give us the efficient level of a market variable directly. Computing such formula is actually simple.

10.4.1. V-u ratio

Our first step in improving the efficiency formula is to introduce the v-u ratio, v/u . This is the ratio of the visit rate to the slack rate, which is also the ratio of the number of visits to the number of unsold goods. Furthermore, in the dynamic model of chapter 9, the v-u ratio coincides with the market tightness, $v/u = \theta$.

In general the market tightness is the ratio of the number of visits to the number of goods for sale in the matching function. In static models the number of goods for sale is the market capacity, while in dynamic models the number of goods for sale is the number of goods currently unsold, which is why the v-u ratio corresponds to the market tightness only in dynamic models. In any case, the rest of the section is about the v-u ratio.

10.4.2. Beveridge elasticity

Our second step is to introduce the Beveridge elasticity, which we denote β . The Beveridge elasticity is defined by

$$\beta = -\frac{u}{v} \cdot v'(u).$$

Hence the Beveridge elasticity is the elasticity of the Beveridge curve, normalized to be positive by multiplying it by -1 . (Recall that the Beveridge curve is downward sloping so $v'(u) < 0$.)

10.4.3. Sufficient-statistic formula

To obtain a sufficient-statistic formula, we rework the efficiency condition (10.3). We start by multiplying both sides of (10.3) by $-(v/u)(u/v) = -1$. We obtain:

$$\frac{v}{u} \cdot \left[-\frac{u}{v} \cdot v'(u) \right] = \frac{1}{\kappa}.$$

The left-hand side is just the v-u ratio times the Beveridge elasticity, β . Dividing both sides by β , we therefore get:

$$(10.5) \quad \left(\frac{v}{u} \right)^* = \frac{1}{\beta \kappa}.$$

Equation (10.5) is our sufficient-statistic formula for market efficiency. It says that the efficient v-u ratio is determined by two sufficient statistics. The first sufficient statistic is the matching cost, κ . The key comparative static is that if the matching cost increases, then $(v/u)^*$ falls and u^* increases. If you have a higher matching cost, it is much more costly to visit shops, so the planner prefers to leave more goods unsold so fewer goods are devoted to matching and in the end more goods are consumed.

The second sufficient statistic is the Beveridge elasticity, β . This statistic matters because the market planner is trying to solve the tradeoff between visits and slack. When the elasticity of the Beveridge curve is high, it is costly to cut slack a bit, in the sense that a small reduction in slack requires many visits. Conversely, by increasing just a bit the slack rate, you can reduce the number of visits considerably. In other words, slack is more favorable when the Beveridge curve is steeper: a higher Beveridge elasticity leads to a lower $(v/u)^*$, so a slacker market.

This formula for market efficiency is quite simple given that it only involves two statistics. If a government wants to know the efficient v-u ratio in any market, they just need to know the Beveridge elasticity and matching cost in that market to determine the efficient market allocation.

The efficient v-u ratio is depicted in figure 10.1D, together with a v-u ratio that is inefficient. In the diagram, the v-u ratio is represented by the slope of any ray through the origin. The efficient slack and visit rates are then given by the intersection of the Beveridge curve with the ray whose slope is given by $(v/u)^*$.

10.5. Applying the formula to the static market

Formula (10.5) expresses the efficient v-u ratio as a function of two sufficient statistics. It is natural to ask how the efficiency condition would look in specific models. The sufficient statistics do not necessarily correspond to parameters from the models, so it is unclear how efficiency might look. In the rest of the chapter we show how the formula applies to all the models presented so far in the book. In this section we apply the formula to the basic static model of chapter 5.

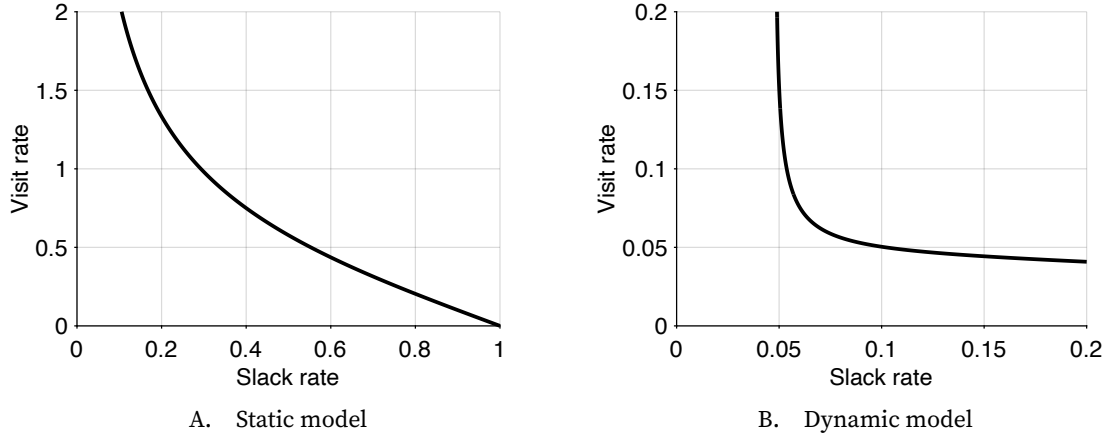


FIGURE 10.2. Beveridge curve in slackish market models

In the static model, the Beveridge curve is given by (10.6). In the dynamic model, the Beveridge curve is given by (9.5). The matching function is CES, given by (4.11). Parameters are set to $\sigma = 2$ and $\lambda = 5\%$.

10.5.1. Verifying the sufficient-statistic assumptions

We need to check that the static model satisfies all the assumptions made in the sufficient-statistic analysis.

First, welfare is solely determined by the consumption of buyers. Assuming that all buyers are the same to avoid distributional issues, welfare is solely determined by aggregate consumption. This in turn is just the number of goods for sale k minus unsold goods uk minus goods used for matching $\kappa v k$. Hence, welfare is proportional to $1 - u - \kappa v$, just as in the sufficient-statistic analysis.

Second, the model features a Beveridge curve. The slack rate is given by $u = 1 - f(\theta)$. But in the static model the market tightness θ is just equal to the visit rate v , so the equation relates visit rate and slack rate:

$$(10.6) \quad u = 1 - f(v).$$

Given that the selling probability f is strictly increasing and concave, we learn that the slack rate u is a strictly decreasing and convex function of the visit rate v . Equivalently, the visit rate v is a strictly decreasing and convex function of the slack rate u (result G.19). Thus, the model admits a Beveridge curve—illustrated in figure 10.2A.

10.5.2. Applying the formula

All the sufficient-statistic conditions are satisfied, which tells us that formula (10.5) holds. We now transform the sufficient-statistic formula into a structural formula, which links the efficient market tightness θ^* to parameters and functions of the static model.

The main step is to compute the Beveridge elasticity in the static model. Equation (10.6) implies that the elasticity of u with respect to v is

$$\epsilon_v^u = -\frac{f(\theta)}{1-f(\theta)} \cdot [1-\eta(\theta)],$$

where $\eta(\theta)$ is the matching elasticity, so $1-\eta(\theta)$ is the elasticity of $f(\theta)$ with respect to θ . The Beveridge elasticity is defined by $\beta = -\epsilon_u^v = -1/\epsilon_v^u$. Therefore, the Beveridge elasticity relates to the market tightness as follows:

$$(10.7) \quad \beta(\theta) = \frac{1}{1-\eta(\theta)} \cdot \frac{u(\theta)}{1-u(\theta)}.$$

Given the matching elasticity $\eta(\theta)$ and slack rate $u(\theta)$ are determined by the matching function, the Beveridge elasticity is solely determined by the matching function.

We then rapidly apply formula (10.5) to the static model. Since the visit rate equals market tightness in that model, the formula imposes

$$\theta^* \beta(\theta^*) \kappa = u(\theta^*),$$

where the Beveridge elasticity $\beta(\theta)$ is given by (10.7) and the slack rate $u(\theta)$ by (5.4). As both $\beta(\theta)$ and $u(\theta)$ are solely determined by the matching function in this model, we have obtained a structural efficiency condition that links the efficient tightness θ^* to the matching function and the matching cost κ .

10.5.3. Hosios condition

By looking closely, we see an interesting property of formula (10.5.2): it implies the famous Hosios (1990) condition. The Hosios condition says that in a market organized around a matching function, and with bargained prices, efficiency is achieved when the seller's bargaining power equals the matching elasticity:

$$\chi = \eta.$$

It turns out that this is an implication of formula (10.5.2).

In the static model under bargaining, we saw with equation (7.7) that the matching wedge satisfies $\tau(\theta) = (1-\chi)/\chi$.

Formula (10.5.2) can be rewritten in a comparable way. Using the expression for the Beveridge elasticity, given by (10.7), we rewrite the efficiency condition (10.5.2) as

$$\kappa \theta^* = [1-\eta(\theta^*)] [1-u(\theta^*)].$$

Then, using the expressions of the matching wedge and slack rate, given by (5.8) and (5.4), we know that

$$\frac{\tau(\theta)}{1 + \tau(\theta)} \cdot [1 - u(\theta)] = \kappa\theta.$$

Combining these last two equations, we infer that the efficient tightness given by (10.5.2) also satisfies

$$(10.8) \quad \tau(\theta^*) = \frac{1 - \eta(\theta^*)}{\eta(\theta^*)}.$$

Hence, for the market to operate efficiently under bargaining, it must be that $\tau(\theta) = (1 - \chi)/\chi = [1 - \eta(\theta^*)]/\eta(\theta^*)$, which requires that $\chi = \eta(\theta^*)$ —which is just what the Hosios condition requires.

In section 7.3, we saw that the price obtained through bargaining equals a fraction χ of the marginal utility of consumption. From this we infer that the static slackish market model operates efficiently if the market price equals a share $\eta(\theta)$ of the marginal utility of consumption. Equivalently, the static market operates efficiently if the market price allocates to the seller a fraction $\eta(\theta)$ of the marginal utility of consumption enjoyed by the buyer.

10.6. Applying the formula to the market with endogenous capacity

The sufficient-statistic analysis did not mention endogenous capacity, so it is natural to worry that the formula might not apply with endogenous market participation. In this section, we show that this worry is not warranted: by an envelope argument, the sufficient-statistic formula continues to apply even when market participation is endogenous, as in the model of chapter 8.

10.6.1. Validity of the formula with endogenous capacity

Is formula (10.5) still valid when sellers enter endogenously? We now turn to the welfare function in this generalized framework. Note first that we do not need to take the utility from money into account since the stock of money is fixed and money just changes hands between buyers and sellers, who value it similarly. So we only need to take into account utility from consumption and disutility from participation.

Market welfare is the sum of individual utilities for all buyers and sellers. All buyers are the same by assumption (to avoid distributional issues) so their utility is just the utility from aggregate consumption: $\mathcal{B}(c(u)) = ac(u)^{1-\alpha}$. Aggregate consumption c is a function of the unemployment rate u given by

$$c(u) = [1 - u - \kappa v(u)] kh(u),$$

where $v(u)$ is the Beveridge curve. Aggregate consumption is obtained by subtracting matching costs $\kappa vkh(u)$ from output $(1-u)kh(u)$.

All sellers have different utility from nonparticipation, so the aggregate utility is obtained by integrating the individual utilities of all potential sellers who remain outside of the market: $\mathcal{S}(h(u)) = \int_{h(u)}^1 \xi i^\phi di$, where the participation rate $h(u)$ comes from (8.2).

Overall, then, market welfare is a function of the slack rate: $\mathcal{W}(u) = \mathcal{B}(c(u)) + \mathcal{S}(h(u))$. The planner cannot do anything about matching on the market, so it takes the Beveridge curve as given. It cannot force people to enter the market or leave it, so it must respect people's voluntary entry or exit of the market, described by (8.2). What the planner can do is influence prices, for instance through taxes or subsidies or price caps, or purchase goods directly on the market. Through these actions it can determine the slack rate u .

The planner chooses the slack rate u to maximize market welfare $\mathcal{W}(u)$. The first-order condition for the maximization problem is $0 = \mathcal{W}'(u)$, which becomes:

$$0 = \mathcal{B}'(c) \cdot \{[-1 - \kappa v'(u)]kh(u) + [1 - u - \kappa v(u)]kh'(u)\} + \mathcal{S}'(h)h'(u).$$

Rearranging terms, we get

$$0 = \mathcal{B}'(c)kh(u) \cdot [-1 - \kappa v'(u)] + \frac{h'(u)}{h(u)} \cdot [\mathcal{B}'(c)c(u) + \mathcal{S}'(h)h(u)].$$

We now need to rework the second term in the first-order condition. Equation (5.13) tells us that

$$\mathcal{B}'(c) = a(1 - \alpha)c^{-\alpha} = p[1 + \tau(\theta)] \quad \text{so} \quad \mathcal{B}'(c)c = p[1 + \tau(\theta)]c = py.$$

By Leibniz's integral rule, we see that $\mathcal{S}'(h) = -\xi h^\phi$ (result G.6). Then, using equation (8.2), we learn that

$$\mathcal{S}'(h) = -p(1 - u)k \quad \text{so} \quad \mathcal{S}'(h)h = -p(1 - u)kh = -py.$$

Hence, we have just established that

$$\mathcal{B}'(c)c(u) + \mathcal{S}'(h)h(u) = 0.$$

This result is key, and it hinges on an envelope-theorem logic. Because market participants behave optimally, they enter until the marginal cost from entering (foregone leisure) is just equal to the marginal benefit from entering, which is determined by the price of goods. But the price of goods also determines the marginal utility of consumption given that buyers behave optimally. All in all, when the planner changes the slack rate, it does affect participation, but this change in participation has no effect on welfare because participation has been chosen optimally by sellers.

Critically, we have just learned that the second term in the first-order equation is just 0. Therefore, the first-order condition reduces to $v'(u) = -1/\kappa$, just as in the baseline case with fixed participation. We therefore recover the result that the efficient v-u ratio is given by $(v/u)^* = 1/(\beta\kappa)$.

In sum, endogenizing market participation does not change the welfare analysis. This result stems from an envelope-theorem logic that is classic in public economics. Even if the market planner alters market participation by changing the slack rate, welfare is unaffected because the sellers who move in or out of the market are indifferent between participating or not. Indeed, if sellers strictly preferred participation to nonparticipation, they would move into the market. Conversely, if they strictly preferred nonparticipation, they would move out of the market.

10.6.2. Applying the formula

In the static market with endogenous capacity, the welfare function is different because it also includes the welfare of sellers, who decide whether to enter the market or not. What makes this model all the more intriguing is that the planner takes into account the fact that it can affect the number of sellers in the market by picking the tightness of the market, as equation (8.2) shows. Despite these differences, we have seen that the v-u ratio remains given by the sufficient-statistic formula (10.5) in this model.

How does the sufficient-statistic formula (10.5) look in the model with endogenous capacity? Just like in the model with exogenous capacity, the Beveridge curve is obtained from the definition of the slack rate— $u = 1 - f(v)$ —so the sufficient-statistic formula translates to the structural formula (10.5.2).

10.7. Applying the formula to the dynamic market

Finally, we apply the sufficient-statistic formula to the dynamic model of chapter 9. Although we did not mention any dynamics in the welfare analysis, the formula remains valid here because the dynamic model features a Beveridge curve too.

10.7.1. Verifying the sufficient-statistic assumptions

We now check that our dynamic model satisfies all the assumptions made in this chapter's welfare analysis. First, welfare is solely determined by the consumption of buyers, just as in the static case.

Second, the model features a Beveridge curve, given implicitly by (9.5). In chapter 9 we established that the Beveridge curve $v(u)$ is downward sloping but we did not check its convexity, which is key for the welfare analysis. We can do that here. We need to

differentiate twice (9.5). The first implicit differentiation gives (9.6). We now implicitly differentiate that equation again with respect to u and obtain

$$0 = \frac{\partial^2 m}{\partial u^2} + 2 \cdot \frac{\partial^2 m}{\partial u \partial v} \cdot v'(u) + \frac{\partial^2 m}{\partial v^2} \cdot [v'(u)]^2 + \frac{\partial m}{\partial v} \cdot v''(u).$$

Reshuffling the terms to isolate the second derivative of the Beveridge curve, we obtain:

$$v''(u) = \frac{-\frac{\partial^2 m}{\partial u^2} - 2 \cdot \frac{\partial^2 m}{\partial u \partial v} \cdot v'(u) - \frac{\partial^2 m}{\partial v^2} \cdot [v'(u)]^2}{\frac{\partial m}{\partial v}}.$$

However, the cross-partial derivative of the matching function is linked to the own-partial derivatives (result C.2):

$$\frac{\partial^2 m}{\partial b \partial s} = \sqrt{\left(-\frac{\partial^2 m}{\partial b^2}\right) \cdot \left(-\frac{\partial^2 m}{\partial s^2}\right)}.$$

Hence, we simplify the expression for the curvature of the Beveridge curve:

$$v''(u) = \frac{\left[\sqrt{-\frac{\partial^2 m}{\partial u^2}} - v'(u) \cdot \sqrt{-\frac{\partial^2 m}{\partial v^2}}\right]^2}{\frac{\partial m}{\partial v}}.$$

Using this expression, we proceed to show that $v''(u) > 0$. Since the matching function is strictly increasing in both arguments, the denominator is strictly positive: $\partial m / \partial v > 0$. Furthermore, we have assumed that the matching function is strictly concave in its second argument, $\partial^2 m / \partial v^2 < 0$, and we have shown in result C.1 that the matching function is strictly concave in its first argument, $\partial^2 m / \partial u^2 < 0$. We have also shown with (9.7) that the Beveridge curve is strictly decreasing: $v'(u) < 0$. This means that all the elements in the square are strictly positive: $\sqrt{-\partial^2 m / \partial v^2} > 0$, $-v'(u) > 0$, and $\sqrt{-\partial^2 m / \partial u^2} > 0$. Hence the square is strictly positive, and so is the whole expression: $v''(u) > 0$. Thus, the Beveridge curve in the dynamic model is strictly convex.

As an illustration, figure 10.2B displays the Beveridge curve obtained in the dynamic model under our usual calibration. Just like the market supply, the Beveridge curve is much more convex in the dynamic model than in the static model.

10.7.2. Applying the formula

All the sufficient-statistic conditions are satisfied, so formula (10.5) holds. We now transform the sufficient-statistic formula into a structural formula, which links the efficient market tightness θ^* to parameters and functions of the dynamic model.

The key step is to compute the Beveridge elasticity in the dynamic model. From the balanced-flow equation, $m(u, v) = \lambda(1 - u)$, which implicitly defines the Beveridge curve,

we computed the slope of the Beveridge curve, and found that it was given by equation (9.7). We now rework this equation to compute the Beveridge elasticity. We first multiply both sides of equation (9.7) by $-u/v$:

$$-\frac{u}{v}v'(u) = \frac{\lambda \cdot \frac{u}{m(u,v)} + \frac{u}{m(u,v)} \cdot \frac{\partial m}{\partial u}}{\frac{v}{m(u,v)} \cdot \frac{\partial m}{\partial v}}.$$

We divided numerator and denominator of the right-hand side fraction by the number of matches $m(u, v)$ to make the elasticities of the matching function appear.

We now simplify this expression. The left-hand side is just the Beveridge elasticity β , by definition. The second term in the right-hand numerator is the matching elasticity, η . From the balanced-flow equation, $m(u, v)/\lambda = (1 - u)$, so the first term in the right-hand numerator boils down to $u/(1 - u)$. The right-hand denominator is the elasticity of the matching function with respect to the number of visits, which is $1 - \eta$, as we established in (4.3). All in all, we obtain the following expression for the Beveridge elasticity:

$$(10.9) \quad \beta(\theta) = \frac{1}{1 - \eta(\theta)} \left[\eta(\theta) + \frac{u(\theta)}{1 - u(\theta)} \right].$$

Just as in the static case, the Beveridge elasticity is a function of tightness that is closely related to the matching elasticity, $\eta(\theta)$. In this dynamic case, the Beveridge elasticity is determined by the matching function and the separation rate λ .

We see that in the dynamic model, the Beveridge elasticity is likely to be stable even if slack varies over time. Indeed, the slack rate is generally small, so in (10.9) the term $u/(1 - u)$ is much smaller than η , which keeps the elasticity stable. This is different from what happens in the static model, where the Beveridge elasticity varies strongly with slack (equation (10.7)). In the dynamic model, formula (10.5) is therefore more usable than the original formula (10.3) as the statistics that it involves (β , κ) are stable over time and do not need to be constantly re-estimated.

In the dynamic model, the v - u ratio is the market tightness: $\theta = v/u$. Thus formula (10.5) simply gives:

$$(10.10) \quad \theta^* \beta(\theta^*) \kappa = 1,$$

where the Beveridge elasticity is given by (10.9). This is the structural condition for efficiency in the dynamic model. It links the efficient tightness θ^* to the matching function, the separation rate λ , and matching cost κ .

10.7.3. Hosios condition

Here too, just like in the static case, formula (10.10) implies the Hosios (1990) condition.

In the dynamic model under bargaining, we saw with equation (9.19) that the market tightness satisfies $\tau(\theta)/u(\theta) = (1 - \chi)/\chi$, where χ is the seller's bargaining power.

Formula (10.10) can be rewritten in a comparable way. Given the expression of the Beveridge elasticity in (10.9), the formula gives:

$$\kappa\theta^* [(1 - u(\theta^*))\eta + u(\theta^*)] = [1 - \eta(\theta^*)] [1 - u(\theta^*)].$$

From the expressions for the matching wedge and slack rate, given by (9.13) and (9.8), we know that

$$\frac{\tau(\theta)}{1 + \tau(\theta)} \cdot \frac{1 - u(\theta)}{u(\theta)} = \kappa\theta.$$

Hence, the efficiency condition above becomes

$$\frac{\tau(\theta^*)}{u(\theta^*)} [(1 - \eta(\theta^*))u(\theta^*) + \eta(\theta^*)] = [1 - \eta(\theta^*)] [1 + \tau(\theta^*)].$$

Subtracting $[1 - \eta(\theta^*)]\tau(\theta^*)$ from both sides, and dividing both sides by $\eta(\theta^*)$, we finally find that the efficient tightness θ^* is implicitly defined by

$$(10.11) \quad \frac{\tau(\theta^*)}{u(\theta^*)} = \frac{1 - \eta(\theta^*)}{\eta(\theta^*)}.$$

This alternative specification of the efficient tightness in a dynamic slackish model is more cumbersome than (10.10), and it has a less direct link to the Beveridge curve and interpretation, but it is useful here and in various situations in the next chapters.

Just as in the static model, for the market to operate efficiently under bargaining, it must be that $\tau/u = (1 - \chi)/\chi = [1 - \eta(\theta^*)]/\eta(\theta^*)$, which again requires that $\chi = \eta(\theta^*)$ —which is what the Hosios condition requires.

In (9.18), we saw that in the dynamic model the bargained price is set at a markdown $1 + [(1 - \chi)/\chi]u$ below the marginal utility of consumption. From this we infer that the dynamic slackish market model operates efficiently if the market price is set at a markdown

$$1 + \frac{1 - \eta(\theta^*)}{\eta(\theta^*)} \cdot u(\theta^*)$$

below the marginal utility of consumption.

10.8. Modulating the social cost of slack

So far we have assumed that unsold goods are pure waste: they have no social value or cost besides their wastefulness.

This is a valid assumption for services that are not rendered or goods that are perishable. But if, for instance, unsold goods can be donated, these goods would have some social value. In the United States in 2023, retailers donate 12% of donatable unsold food to food banks (ReFED 2025). On the other hand, if unsold goods are sent to landfills where they produce noxious gases while decomposing, these goods impose a social cost. In the United States in 2023, about 25 million tons of surplus food end up in landfill, emitting a sizable amount of methane (ReFED 2025).

Similarly, the assumption seems appropriate for employed workers who are idle. But if an unemployed worker engages in home production, their time might have social value. If on the other hand the time spent unemployed is painful psychologically, their time generates a social cost.

To allow for the possibility that unsold goods have some social value or cost beyond the waste they impose, we introduce a statistic $\zeta < 1$, which captures the social value of unsold goods compared to sold goods. The case $\zeta = 0$, which we have considered so far, is when unsold goods are pure waste. The case $\zeta < 0$ is when unsold goods generate a cost on top of being wasteful. And the case $\zeta \in (0, 1)$ is when unsold goods have some social value that alleviates the wastefulness of unsold goods.

In this extension, the market welfare has an additional element. Since each of the uk unsold goods have a social value ζ relative to sold goods, an amount ζuk must be added to market welfare (10.1). In total, market welfare becomes

$$\mathcal{W}(u) = [1 - \kappa v(u) - (1 - \zeta)u] k.$$

Given that the welfare cost of a good failing to sell is reduced by $1 - \zeta$, the first-order condition (10.3) generalizes to

$$(10.12) \quad v'(u) = -\frac{1 - \zeta}{\kappa}.$$

If slack has some social value ($\zeta > 0$), the isowelfare curve is flatter in the efficiency diagram, so the tangency point occurs further out, at a higher efficient slack rate (just like in figure 10.1C). Along the Beveridge curve, the efficient slack rate u^* increases. Intuitively, when the cost of having unsold goods is a bit less, the efficient allocation of goods tolerates a bit more slack. Conversely, if slack imposes a social cost beyond its wastefulness ($\zeta < 0$), the isowelfare curve is steeper, so the efficient slack rate is lower.

Finally, the sufficient-statistic formula (10.5) generalizes to

$$(10.13) \quad \left(\frac{v}{u}\right)^* = \frac{1 - \zeta}{\beta \kappa}.$$

In this generalized formula, a third sufficient statistic appears: the social value of unsold goods, ζ . The efficient v - u ratio responds to changes in the social value of unsold goods just like to changes in the matching cost κ : when the matching cost increases or when the social value increases, the efficient v - u ratio decreases, so the efficient slack rate increases.

10.9. Are markets efficient?

The last question we ask in this chapter is: Can we expect markets to operate efficiently? One of the key theorems of Walrasian theory is that markets operate efficiently. But slackish markets are different: there is absolutely no guarantee that a slackish market operates efficiently.

From a theoretical perspective, the key reason why Walrasian markets are efficient while slackish markets are not is that prices are set differently. In a slackish market, there exists a price norm that guarantees efficiency, but this price norm might not be adopted in the real world. Under all other price norms, efficiency is not guaranteed. In effect, there is no guarantee that the price norm that prevails in the real world leads to efficiency.

In any slackish market, the market tightness is given by the supply-equals-demand condition. When we solve the model, we use the price given by the price norm, plug it into the market demand, and determine the tightness that equalizes supply and demand.

Here we use the condition in the other way: we set tightness to its efficient level θ^* . Then we use the supply-equals-demand condition to find the efficient price p^* , defined by

$$y^d(\theta^*, p^*) = y^s(\theta^*).$$

We know that the market demand is strictly decreasing in the price, so that equation defines a unique efficient price p^* . If the price norm happens to set the price to p^* , the market will operate efficiently. But there is no guarantee that this occurs in reality, as the formation of norms is a complex process that is not geared toward efficiency.

Furthermore, by using the functional forms for market supply and demand, we can obtain an expression for the efficient price. For concreteness, let's find the expression for the efficient price in the basic model of chapter 5. Supply and demand curves are given by (5.15) and (5.5). Hence, the supply-equals-demand condition requires:

$$\left[\frac{(1 - \alpha)a}{p^*} \right]^{1/\alpha} [1 + \tau(\theta^*)]^{1-1/\alpha} = [1 - u(\theta^*)] k.$$

From (10.8), we know that $1 + \tau(\theta^*) = 1/\eta(\theta^*)$. By reshuffling terms, we obtain the following expression for the efficient price:

$$p^* = (1 - \alpha) [1 - u(\theta^*)]^{-\alpha} \eta(\theta^*)^{1-\alpha} \cdot a k^{-\alpha},$$

where the efficient tightness θ^* is equivalently defined by (10.5.2) or (10.8).

An interesting aspect of the efficient price is that it is a flexible case of the price norm (7.1) introduced in chapter 7. Indeed, we obtain the price p^* from the price norm by setting the price-rigidity parameter to $\gamma = 0$ and the price-level parameter to

$$\rho = (1 - \alpha) [1 - u(\theta^*)]^{-\alpha} \eta(\theta^*)^{1-\alpha},$$

where the efficient tightness θ^* is independent of the parameters a and k , as it is defined by (10.5.2).

We argued earlier in the book that bargaining was a possible mechanism pushing prices towards flexibility. We see here that efficiency is another one. Any social or economic mechanism pushing markets towards efficiency makes prices more flexible. In chapter 15 we will introduce such a market mechanism: Moen (1997)'s directed search.

Because market inefficiency is generic, government interventions might be needed to bring markets and the economy closer to efficiency—as we discuss in part V. Here there is no guarantee that Smith's invisible hand maintains markets at efficiency. The reason is that with slack, the hand is unable to influence prices and keep them at their efficient level. Hence, there is generally too much or too little slack on any market.

10.10. Summary

In this chapter we define efficiency as the level of slack that maximizes market welfare. Rather than specifying a complete structural model, we adopt a sufficient-statistic approach: We analyze efficiency in any market that admits a Beveridge curve, and express the results in terms of observable statistics.

We find that the efficient slack rate equates the slope of the Beveridge curve to the inverse of the matching cost. From this, we obtained a sufficient-statistic formula that links the efficient v - u ratio to the matching cost κ and Beveridge elasticity β : $(v/u)^* = 1/(\beta\kappa)$.

Finally, we argue that slackish markets are generically inefficient, since no market mechanism ensures that prices adjust to their efficient level. In practice, price norms might generate either excessive or insufficient slack, justifying policy interventions that move markets toward their efficient operating points.

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